

```
python main.py --task train --run_name jump_split --data_split addprim_jump --
n_layer 2 --n_head 2 --n_embd 16 --max_epochs 60 --batch_size 32 --num_workers 8
--learning_rate 4e-4 --max_len 128 --seed 44 --grad_norm_clip 1.0
No GPU available, falling back to CPU
Downloading data: 100%|
| 2.58M/2.58M [00:00<00:00, 6.73MB/s]
Downloading data: 100%|
| 1.53M/1.53M [00:00<00:00, 10.6MB/s]
Generating train split: 100%|
| 14670/14670 [00:00<00:00, 38915.85 examples/s]
Generating test split: 100%|
| 7706/7706 [00:00<00:00, 53547.12 examples/s]
Building tokenizer at ./tokenizer/addprim_jump_vocab.json.
Building tokenizer for actions: 100%|
| 14670/14670 [00:00<00:00, 30098.02it/s]
Building tokenizer for commands: 100%|
| 14670/14670 [00:00<00:00, 30268.35it/s]
tokenizer saved
{'<pad>': 0, '<s>': 1, '</s>': 2, '<unk>': 3, 'I_JUMP': 4, 'I_TURN_RIGHT': 5,
'I_LOOK': 6, 'I_TURN_LEFT': 7, 'I_WALK': 8, 'I_RUN': 9, 'jump': 10, 'look': 11,
'opposite': 12, 'right': 13, 'thrice': 14, 'and': 15, 'walk': 16, 'left': 17,
'run': 18, 'after': 19, 'twice': 20, 'turn': 21, 'around': 22}
train dataset size: 13203
val dataset size: 1467
loading model
total params: 9408
C:
\\Users\\Chayce\\Documents\\CollegeFinalSemester\\MachineLearning\\HW5\\HW5\\code\\traine
r.py:79: FutureWarning: `torch.cuda.amp.GradScaler(args...)` is deprecated.
Please use `torch.amp.GradScaler('cuda', args...)` instead.
    scaler = GradScaler()
C:\\Users\\Chayce\\AppData\\Roaming\\Python\\Python312\\site-
packages\\torch\\amp\\grad_scaler.py:132: UserWarning: torch.cuda.amp.GradScaler is
enabled, but CUDA is not available. Disabling.
    warnings.warn(
epoch 1 iter 0: train loss 0.00000. lr 0.0000e+00: 0%|
| 0/413 [00:00<?, ?it/s]C
:
\\Users\\Chayce\\Documents\\CollegeFinalSemester\\MachineLearning\\HW5\\HW5\\code\\traine
r.py:102: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated.
Please use `torch.amp.autocast('cuda', args...)` instead.
    with torch.cuda.amp.autocast():
C:\\Users\\Chayce\\AppData\\Roaming\\Python\\Python312\\site-
packages\\torch\\amp\\autocast_mode.py:266: UserWarning: User provided device_type
of 'cuda', but CUDA is not available. Disabling
    warnings.warn(
epoch 1 iter 412: train loss 1.41066. lr 3.9978e-04: 100%|
| 413/413 [00:16<00:00, 24.85it/s]
test loss: %f 1.3522288047749063
epoch_valid_loss: 1.3522288047749063, epoch_train_loss: 2.031510356552087,
epoch: 1
Saving at epoch 1:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 1.1333833932876587 train_step: 500, learning_rate:
0.000399664942057356
86/413 [00:03<00:12, 26.94it/s]
epoch 2 iter 412: train loss 0.88993. lr 3.9902e-04: 100%|
```

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| 413/413 [00:16<00:00, 25.05it/s]
test loss: %f 0.794186031040938
epoch_valid_loss: 0.794186031040938, epoch_train_loss: 1.0417945309932237,
epoch: 2
Saving at epoch 2:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.7617142200469971 train_step: 1000, learning_rate:
0.00039854005329423314
173/413 [00:06<00:08, 27.23it/s]
epoch 3 iter 412: train loss 0.72993. lr 3.9773e-04: 100%|
```

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| 413/413 [00:16<00:00, 25.25it/s]
test loss: %f 0.6714309415091639
epoch_valid_loss: 0.6714309415091639, epoch_train_loss: 0.799130122684682,
epoch: 3
Saving at epoch 3:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.6928979754447937 train_step: 1500, learning_rate:
0.0003966271561634415
260/413 [00:09<00:05, 27.70it/s]
epoch 4 iter 412: train loss 0.68271. lr 3.9590e-04: 100%|
```

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| 413/413 [00:16<00:00, 25.28it/s]
test loss: %f 0.6079351765953976
epoch_valid_loss: 0.6079351765953976, epoch_train_loss: 0.730230902355462,
epoch: 4
Saving at epoch 4:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.6557028889656067 train_step: 2000, learning_rate:
0.0003939338503183291
347/413 [00:12<00:02, 27.45it/s]
epoch 5 iter 412: train loss 0.58654. lr 3.9354e-04: 100%|
```

```
| 413/413 [00:16<00:00, 25.41it/s]
test loss: %f 0.5832546657842138
epoch_valid_loss: 0.5832546657842138, epoch_train_loss: 0.6889764375028541,
epoch: 5
Saving at epoch 5:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
epoch 6 iter 412: train loss 0.61355. lr 3.9065e-04: 100%|
```

```
| 413/413 [00:16<00:00, 25.16it/s]
test loss: %f 0.5472460609415303
epoch_valid_loss: 0.5472460609415303, epoch_train_loss: 0.6634222075667855,
epoch: 6
Saving at epoch 6:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.6789405345916748 train_step: 2500, learning_rate:
0.0003904739496599693
20/413 [00:00<00:16, 23.81it/s]
epoch 7 iter 412: train loss 0.65060. lr 3.8725e-04: 100%|
```

```
| 413/413 [00:16<00:00, 25.43it/s]
test loss: %f 0.5342596335255582
epoch_valid_loss: 0.5342596335255582, epoch_train_loss: 0.6464074633600637,
epoch: 7
Saving at epoch 7:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.6719090342521667 train_step: 3000, learning_rate:
0.0003862555585368648
107/413 [00:04<00:11, 27.44it/s]
epoch 8 iter 412: train loss 0.67747. lr 3.8334e-04: 100%|
```

```
██████████ | 413/413 [00:16<00:00, 25.42it/s]
test loss: %f 0.5064147730236468
epoch_valid_loss: 0.5064147730236468, epoch_train_loss: 0.6290081976112384,
epoch: 8
Saving at epoch 8:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.5223168134689331 train_step: 3500, learning_rate:
0.0003812979233347176 ██████████ |
194/413 [00:07<00:07, 27.93it/s]
epoch 9 iter 412: train loss 0.65615. lr 3.7893e-04: 100%|
```

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██████████ | 413/413 [00:16<00:00, 25.35it/s]
test loss: %f 0.4914279275614282
epoch_valid_loss: 0.4914279275614282, epoch_train_loss: 0.6162610066283413,
epoch: 9
Saving at epoch 9:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.6501168012619019 train_step: 4000, learning_rate:
0.00037562072079497927 ██████████ |
283/413 [00:10<00:04, 27.50it/s]
epoch 10 iter 412: train loss 0.69187. lr 3.7405e-04: 100%|
```

```
██████████ | 413/413 [00:16<00:00, 25.43it/s]
test loss: %f 0.46965566795805225
epoch_valid_loss: 0.46965566795805225, epoch_train_loss: 0.6041386223445504,
epoch: 10
Saving at epoch 10:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.6576071977615356 train_step: 4500, learning_rate:
0.0003692464836055114 ██████████ |
368/413 [00:13<00:01, 27.51it/s]
epoch 11 iter 412: train loss 0.74595. lr 3.6869e-04: 100%|
```

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██████████ | 413/413 [00:16<00:00, 25.30it/s]
test loss: %f 0.4638578081908433
epoch_valid_loss: 0.4638578081908433, epoch_train_loss: 0.5925910050008834,
epoch: 11
Saving at epoch 11:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
epoch 12 iter 412: train loss 0.66672. lr 3.6287e-04: 100%|
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██████████ | 413/413 [00:16<00:00, 24.44it/s]
test loss: %f 0.4420263190632281
epoch_valid_loss: 0.4420263190632281, epoch_train_loss: 0.58008401445846, epoch:
12
Saving at epoch 12:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.5890619158744812 train_step: 5000, learning_rate:
0.0003622065059745662 ██████████ |
44/413 [00:01<00:14, 24.66it/s]
epoch 13 iter 412: train loss 0.49256. lr 3.5661e-04: 100%|
```

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██████████ | 413/413 [00:16<00:00, 25.37it/s]
test loss: %f 0.4386257062787595
epoch_valid_loss: 0.4386257062787595, epoch_train_loss: 0.570319776240619,
epoch: 13
Saving at epoch 13:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.5898802876472473 train_step: 5500, learning_rate:
0.00035451727856479693 ██████████ |
131/413 [00:05<00:09, 28.31it/s]
epoch 14 iter 412: train loss 0.64943. lr 3.4993e-04: 100%|
```

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██████████ | 413/413 [00:16<00:00, 25.57it/s]
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test loss: %f 0.4265173420957897
epoch_valid_loss: 0.4265173420957897, epoch_train_loss: 0.5593199990419152,
epoch: 14
Saving at epoch 14:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.5685380697250366 train_step: 6000, learning_rate:
0.0003462147677219963
218/413 [00:08<00:07, 26.86it/s]
epoch 15 iter 412: train loss 0.64623. lr 3.4284e-04: 100%|
| 413/413 [00:16<00:00, 25.33it/s]
test loss: %f 0.4086618177268816
epoch_valid_loss: 0.4086618177268816, epoch_train_loss: 0.5506598017839196,
epoch: 15
Saving at epoch 15:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.47740691900253296 train_step: 6500, learning_rate:
0.0003373319332174821
305/413 [00:11<00:03, 27.93it/s]
epoch 16 iter 412: train loss 0.52655. lr 3.3536e-04: 100%|
| 413/413 [00:16<00:00, 25.45it/s]
test loss: %f 0.40799939308477484
epoch_valid_loss: 0.40799939308477484, epoch_train_loss: 0.5400579004541725,
epoch: 16
Saving at epoch 16:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.49540743231773376 train_step: 7000, learning_rate:
0.00032790404039756746
390/413 [00:14<00:00, 27.86it/s]
epoch 17 iter 412: train loss 0.60357. lr 3.2752e-04: 100%|
| 413/413 [00:16<00:00, 25.27it/s]
test loss: %f 0.41461472925932513
epoch_valid_loss: 0.41461472925932513, epoch_train_loss: 0.5326640766123016,
epoch: 17
epoch 18 iter 412: train loss 0.51466. lr 3.1934e-04: 100%|
| 413/413 [00:16<00:00, 25.33it/s]
test loss: %f 0.39191359475902887
epoch_valid_loss: 0.39191359475902887, epoch_train_loss: 0.5248554672225046,
epoch: 18
Saving at epoch 18:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.5124791860580444 train_step: 7500, learning_rate:
0.0003179767697173972
66/413 [00:02<00:14, 23.90it/s]
epoch 19 iter 412: train loss 0.57030. lr 3.1083e-04: 100%|
| 413/413 [00:16<00:00, 24.71it/s]
test loss: %f 0.3935422340165014
epoch_valid_loss: 0.3935422340165014, epoch_train_loss: 0.5180157233237066,
epoch: 19
step_train_loss: 0.4581458866596222 train_step: 8000, learning_rate:
0.0003075733688473864
152/413 [00:05<00:09, 27.84it/s]
epoch 20 iter 412: train loss 0.51673. lr 3.0202e-04: 100%|
| 413/413 [00:16<00:00, 25.08it/s]
test loss: %f 0.37509436387082806
epoch_valid_loss: 0.37509436387082806, epoch_train_loss: 0.5110751435918323,
epoch: 20
Saving at epoch 20:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
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step_train_loss: 0.5940280556678772 train_step: 8500, learning_rate:
0.00029674303303020035
239/413 [00:09<00:07, 22.86it/s]
epoch 21 iter 412: train loss 0.55416. lr 2.9293e-04: 100%|
| 413/413 [00:17<00:00, 23.20it/s]
test loss: %f 0.36341058624827344
epoch_valid_loss: 0.36341058624827344, epoch_train_loss: 0.4994190875034933,
epoch: 21
Saving at epoch 21:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.531243085861206 train_step: 9000, learning_rate:
0.00028552871622647814
326/413 [00:12<00:03, 27.44it/s]
epoch 22 iter 412: train loss 0.54625. lr 2.8359e-04: 100%|
| 413/413 [00:16<00:00, 25.03it/s]
test loss: %f 0.34742255897625635
epoch_valid_loss: 0.34742255897625635, epoch_train_loss: 0.4906464040856673,
epoch: 22
Saving at epoch 22:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
epoch 23 iter 412: train loss 0.50860. lr 2.7403e-04: 100%|
| 413/413 [00:16<00:00, 25.39it/s]
test loss: %f 0.35364692755367444
epoch_valid_loss: 0.35364692755367444, epoch_train_loss: 0.48471905236960033,
epoch: 23
step_train_loss: 0.5496043562889099 train_step: 9500, learning_rate:
0.0002739844415847242
| 0/413 [00:00<?, ?it/s]
epoch 24 iter 412: train loss 0.44974. lr 2.6427e-04: 100%|
| 413/413 [00:16<00:00, 25.30it/s]
test loss: %f 0.33797925062801526
epoch_valid_loss: 0.33797925062801526, epoch_train_loss: 0.47837538616709213,
epoch: 24
Saving at epoch 24:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.4990847706794739 train_step: 10000, learning_rate:
0.00026213727841585267
86/413 [00:03<00:11, 27.72it/s]
epoch 25 iter 412: train loss 0.57195. lr 2.5433e-04: 100%|
| 413/413 [00:16<00:00, 25.45it/s]
test loss: %f 0.3360501618488975
epoch_valid_loss: 0.3360501618488975, epoch_train_loss: 0.4735336785743658,
epoch: 25
Saving at epoch 25:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.4679524004459381 train_step: 10500, learning_rate:
0.00025004349404391703
175/413 [00:06<00:08, 28.02it/s]
epoch 26 iter 412: train loss 0.41751. lr 2.4424e-04: 100%|
| 413/413 [00:16<00:00, 25.50it/s]
test loss: %f 0.3405913574540097
epoch_valid_loss: 0.3405913574540097, epoch_train_loss: 0.4671829590352915,
epoch: 26
step_train_loss: 0.5518253445625305 train_step: 11000, learning_rate:
0.00023775108837698295
260/413 [00:09<00:05, 27.46it/s]
epoch 27 iter 412: train loss 0.59478. lr 2.3404e-04: 100%|
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██████████ | 413/413 [00:16<00:00, 25.55it/s]
test loss: %f 0.31950214764346246
epoch_valid_loss: 0.31950214764346246, epoch_train_loss: 0.462340919576026,
epoch: 27
Saving at epoch 27:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.4425722658634186 train_step: 11500, learning_rate:
0.00022530882501797897 ██████████ |
347/413 [00:12<00:02, 27.78it/s]
epoch 28 iter 412: train loss 0.36592. lr 2.2374e-04: 100%|
```

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██████████ | 413/413 [00:16<00:00, 25.43it/s]
test loss: %f 0.31685349895902304
epoch_valid_loss: 0.31685349895902304, epoch_train_loss: 0.4548256896598576,
epoch: 28
Saving at epoch 28:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
epoch 29 iter 412: train loss 0.49876. lr 2.1338e-04: 100%|
```

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██████████ | 413/413 [00:16<00:00, 25.64it/s]
test loss: %f 0.31382313759430597
epoch_valid_loss: 0.31382313759430597, epoch_train_loss: 0.45254159753028184,
epoch: 29
Saving at epoch 29:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.4252122640609741 train_step: 12000, learning_rate:
0.00021277636875829337 ██████████ |
23/413 [00:00<00:16, 24.20it/s]
epoch 30 iter 412: train loss 0.39044. lr 2.0298e-04: 100%|
```

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██████████ | 413/413 [00:16<00:00, 25.43it/s]
test loss: %f 0.30880193962999014
epoch_valid_loss: 0.30880193962999014, epoch_train_loss: 0.44593343297448057,
epoch: 30
Saving at epoch 30:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.5463643670082092 train_step: 12500, learning_rate:
0.00020018300736951642 ██████████ |
110/413 [00:04<00:10, 27.85it/s]
epoch 31 iter 412: train loss 0.36874. lr 1.9258e-04: 100%|
```

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██████████ | 413/413 [00:16<00:00, 25.39it/s]
test loss: %f 0.3018396250579668
epoch_valid_loss: 0.3018396250579668, epoch_train_loss: 0.44476848775768973,
epoch: 31
Saving at epoch 31:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.44779160618782043 train_step: 13000, learning_rate:
0.000187588944406008 ██████████ |
197/413 [00:07<00:09, 23.23it/s]
epoch 32 iter 412: train loss 0.41807. lr 1.8219e-04: 100%|
```

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██████████ | 413/413 [00:16<00:00, 25.16it/s]
test loss: %f 0.2955654700813086
epoch_valid_loss: 0.2955654700813086, epoch_train_loss: 0.4381289888380803,
epoch: 32
Saving at epoch 32:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.45385029911994934 train_step: 13500, learning_rate:
0.00017504414059159524 ██████████ |
282/413 [00:10<00:04, 27.48it/s]
epoch 33 iter 412: train loss 0.47014. lr 1.7185e-04: 100%|
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██████████ | 413/413 [00:16<00:00, 25.35it/s]
```

```
test loss: %f 0.29456734495318454
epoch_valid_loss: 0.29456734495318454, epoch_train_loss: 0.4330199863085158,
epoch: 33
Saving at epoch 33:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.3617713749408722 train_step: 14000, learning_rate:
0.00016259838591849837
370/413 [00:13<00:01, 26.94it/s]
epoch 34 iter 412: train loss 0.55475. lr 1.6159e-04: 100%|
| 413/413 [00:16<00:00, 25.35it/s]
test loss: %f 0.2850846644976865
epoch_valid_loss: 0.2850846644976865, epoch_train_loss: 0.4279457415853228,
epoch: 34
Saving at epoch 34:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
epoch 35 iter 412: train loss 0.39602. lr 1.5144e-04: 100%|
| 413/413 [00:16<00:00, 25.11it/s]
test loss: %f 0.2822332210514856
epoch_valid_loss: 0.2822332210514856, epoch_train_loss: 0.4256126903015534,
epoch: 35
Saving at epoch 35:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.49263158440589905 train_step: 14500, learning_rate:
0.00015031098534443494
44/413 [00:01<00:14, 24.89it/s]
epoch 36 iter 412: train loss 0.39873. lr 1.4141e-04: 100%|
| 413/413 [00:16<00:00, 25.36it/s]
test loss: %f 0.2877845841905345
epoch_valid_loss: 0.2877845841905345, epoch_train_loss: 0.421488668886859,
epoch: 36
step_train_loss: 0.4099152386188507 train_step: 15000, learning_rate:
0.00013821075068104198
131/413 [00:05<00:10, 27.55it/s]
epoch 37 iter 412: train loss 0.38533. lr 1.3154e-04: 100%|
| 413/413 [00:16<00:00, 25.44it/s]
test loss: %f 0.27994256434233294
epoch_valid_loss: 0.27994256434233294, epoch_train_loss: 0.41676252055687707,
epoch: 37
Saving at epoch 37:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.39495792984962463 train_step: 15500, learning_rate:
0.000126355755902497
218/413 [00:08<00:07, 27.63it/s]
epoch 38 iter 412: train loss 0.33011. lr 1.2186e-04: 100%|
| 413/413 [00:16<00:00, 25.40it/s]
test loss: %f 0.2684429453118988
epoch_valid_loss: 0.2684429453118988, epoch_train_loss: 0.41423676796167297,
epoch: 38
Saving at epoch 38:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.3858374357223511 train_step: 16000, learning_rate:
0.00011479303070585325
305/413 [00:11<00:04, 26.76it/s]
epoch 39 iter 412: train loss 0.35190. lr 1.1239e-04: 100%|
| 413/413 [00:16<00:00, 25.06it/s]
test loss: %f 0.26945507785548334
epoch_valid_loss: 0.26945507785548334, epoch_train_loss: 0.41312803815121224,
epoch: 39
```

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step_train_loss: 0.32704541087150574 train_step: 16500, learning_rate:
0.00010356851278386925
392/413 [00:14<00:00, 27.60it/s]
epoch 40 iter 412: train loss 0.45021. lr 1.0316e-04: 100%|
| 413/413 [00:16<00:00, 25.41it/s]
test loss: %f 0.26365891034188477
epoch_valid_loss: 0.26365891034188477, epoch_train_loss: 0.40933429199038635,
epoch: 40
Saving at epoch 40:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
epoch 41 iter 412: train loss 0.40435. lr 9.4187e-05: 100%|
| 413/413 [00:16<00:00, 25.50it/s]
test loss: %f 0.2632236992535384
epoch_valid_loss: 0.2632236992535384, epoch_train_loss: 0.40795230216033235,
epoch: 41
Saving at epoch 41:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.39137208461761475 train_step: 17000, learning_rate:
9.273536202219221e-05
65/413 [00:02<00:13, 25.87it/s]
epoch 42 iter 412: train loss 0.37685. lr 8.5503e-05: 100%|
| 413/413 [00:16<00:00, 25.34it/s]
test loss: %f 0.2579678021695303
epoch_valid_loss: 0.2579678021695303, epoch_train_loss: 0.40472955853829373,
epoch: 42
Saving at epoch 42:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.35596033930778503 train_step: 17500, learning_rate:
8.231900034480753e-05
152/413 [00:05<00:09, 26.83it/s]
epoch 43 iter 412: train loss 0.44662. lr 7.7128e-05: 100%|
| 413/413 [00:16<00:00, 25.24it/s]
test loss: %f 0.2587260383626689
epoch_valid_loss: 0.2587260383626689, epoch_train_loss: 0.4015691094190676,
epoch: 43
step_train_loss: 0.29161715507507324 train_step: 18000, learning_rate:
7.236967135128065e-05
239/413 [00:09<00:06, 26.16it/s]
epoch 44 iter 412: train loss 0.44455. lr 6.9086e-05: 100%|
| 413/413 [00:16<00:00, 25.21it/s]
test loss: %f 0.2603056314198867
epoch_valid_loss: 0.2603056314198867, epoch_train_loss: 0.39932554115971985,
epoch: 44
step_train_loss: 0.434173047542572 train_step: 18500, learning_rate:
6.292690572814624e-05
327/413 [00:12<00:03, 27.65it/s]
epoch 45 iter 412: train loss 0.40934. lr 6.1399e-05: 100%|
| 413/413 [00:16<00:00, 25.29it/s]
test loss: %f 0.25189799199933594
epoch_valid_loss: 0.25189799199933594, epoch_train_loss: 0.39899293757235454,
epoch: 45
Saving at epoch 45:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
epoch 46 iter 412: train loss 0.37610. lr 5.4087e-05: 100%|
| 413/413 [00:16<00:00, 25.19it/s]
test loss: %f 0.24714844220358392
epoch_valid_loss: 0.24714844220358392, epoch_train_loss: 0.39401189481375004,
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epoch: 46
Saving at epoch 46:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.4679276943206787 train_step: 19000, learning_rate:
5.403520827610582e-05
2/413 [00:00<00:23, 17.86it/s]
epoch 47 iter 412: train loss 0.32191. lr 4.7170e-05: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████ | 413/413 [00:16<00:00, 25.33it/s]
test loss: %f 0.25345323979854584
epoch_valid_loss: 0.25345323979854584, epoch_train_loss: 0.3948375324360106,
epoch: 47
step_train_loss: 0.4520566165447235 train_step: 19500, learning_rate:
4.5715390077938854e-05
89/413 [00:03<00:13, 24.25it/s]
epoch 48 iter 412: train loss 0.42323. lr 4.0667e-05: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████ | 413/413 [00:16<00:00, 24.48it/s]
test loss: %f 0.24578694156978442
epoch_valid_loss: 0.24578694156978442, epoch_train_loss: 0.39280336554056217,
epoch: 48
Saving at epoch 48:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.3859352767467499 train_step: 20000, learning_rate:
4e-05
174/413 [00:07<00:09, 26.00it/s]
epoch 49 iter 412: train loss 0.47975. lr 4.0000e-05: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████ | 413/413 [00:17<00:00, 24.21it/s]
test loss: %f 0.24902800924104193
epoch_valid_loss: 0.24902800924104193, epoch_train_loss: 0.39274765753139884,
epoch: 49
step_train_loss: 0.43008869886398315 train_step: 20500, learning_rate:
4e-05
263/413 [00:10<00:05, 27.12it/s]
epoch 50 iter 412: train loss 0.30138. lr 4.0000e-05: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████ | 413/413 [00:16<00:00, 24.86it/s]
test loss: %f 0.24657340704098993
epoch_valid_loss: 0.24657340704098993, epoch_train_loss: 0.3905023614228782,
epoch: 50
step_train_loss: 0.40143057703971863 train_step: 21000, learning_rate:
4e-05
348/413 [00:15<00:02, 23.17it/s]
epoch 51 iter 412: train loss 0.39707. lr 4.0000e-05: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████ | 413/413 [00:19<00:00, 21.31it/s]
test loss: %f 0.24492810601773468
epoch_valid_loss: 0.24492810601773468, epoch_train_loss: 0.3906634986472765,
epoch: 51
Saving at epoch 51:
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt
epoch 52 iter 412: train loss 0.43365. lr 4.0000e-05: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████ | 413/413 [00:18<00:00, 22.72it/s]
test loss: %f 0.24726986496344858
epoch_valid_loss: 0.24726986496344858, epoch_train_loss: 0.3886719255845714,
epoch: 52
step_train_loss: 0.4209086000919342 train_step: 21500, learning_rate: 4e-05
| 23/413 [00:01<00:20, 18.82it/s]
epoch 53 iter 412: train loss 0.34005. lr 4.0000e-05: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████ | 413/413 [00:19<00:00, 21.21it/s]
test loss: %f 0.247453977232394
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epoch_valid_loss: 0.247453977232394, epoch_train_loss: 0.3909942580626028,  
epoch: 53  
step_train_loss: 0.39988741278648376 train_step: 22000, learning_rate:  
4e-05  
109/413 [00:04<00:12, 25.19it/s]  
epoch 54 iter 412: train loss 0.43174. lr 4.0000e-05: 100%  
| 413/413 [00:16<00:00, 24.96it/s]  
test loss: %f 0.24578128431154334  
epoch_valid_loss: 0.24578128431154334, epoch_train_loss: 0.39040085435202276,  
epoch: 54  
step_train_loss: 0.3142300844192505 train_step: 22500, learning_rate:  
4e-05  
197/413 [00:07<00:07, 27.13it/s]  
epoch 55 iter 412: train loss 0.38318. lr 4.0000e-05: 100%  
| 413/413 [00:16<00:00, 24.80it/s]  
test loss: %f 0.24282590073087942  
epoch_valid_loss: 0.24282590073087942, epoch_train_loss: 0.38904161293050565,  
epoch: 55  
Saving at epoch 55:  
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt  
step_train_loss: 0.35718634724617004 train_step: 23000, learning_rate:  
4e-05  
284/413 [00:10<00:04, 28.21it/s]  
epoch 56 iter 412: train loss 0.45020. lr 4.0000e-05: 100%  
| 413/413 [00:16<00:00, 25.69it/s]  
test loss: %f 0.2423201537002688  
epoch_valid_loss: 0.2423201537002688, epoch_train_loss: 0.38743259730696966,  
epoch: 56  
Saving at epoch 56:  
./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt  
step_train_loss: 0.4419860541820526 train_step: 23500, learning_rate:  
4e-05  
371/413 [00:14<00:01, 27.83it/s]  
epoch 57 iter 412: train loss 0.38042. lr 4.0000e-05: 100%  
| 413/413 [00:16<00:00, 24.98it/s]  
test loss: %f 0.2449807777352955  
epoch_valid_loss: 0.2449807777352955, epoch_train_loss: 0.3870527968712638,  
epoch: 57  
epoch 58 iter 412: train loss 0.30742. lr 4.0000e-05: 100%  
| 413/413 [00:18<00:00, 22.04it/s]  
test loss: %f 0.24279306083917618  
epoch_valid_loss: 0.24279306083917618, epoch_train_loss: 0.38834499901489833,  
epoch: 58  
step_train_loss: 0.3453693687915802 train_step: 24000, learning_rate: 4e-05  
| 44/413 [00:01<00:15, 24.30it/s]  
epoch 59 iter 412: train loss 0.34860. lr 4.0000e-05: 100%  
| 413/413 [00:16<00:00, 25.19it/s]  
test loss: %f 0.24373750129471655  
epoch_valid_loss: 0.24373750129471655, epoch_train_loss: 0.3863854018236188,  
epoch: 59  
step_train_loss: 0.3549240529537201 train_step: 24500, learning_rate:  
4e-05  
133/413 [00:05<00:11, 25.02it/s]  
epoch 60 iter 412: train loss 0.36386. lr 4.0000e-05: 100%  
| 413/413 [00:17<00:00, 24.15it/s]  
test loss: %f 0.24116106979224994  
epoch_valid_loss: 0.24116106979224994, epoch_train_loss: 0.3859563932436142,
```

epoch: 60

Saving at epoch 60:

./cond_gpt/weights/jump_split_addprim_jumpsplit_2layer_2head_16embd_32bs.pt